

Customer Transaction Prediction

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Abstract—Transaction data is very important in this digital world. Payment service providers' success depend on privacy of user. It is also important to extract the key insight from data to advance in business. We proposed three approaches to solve this issue. We will show comparative analysis for each of the experiment

Keywords—(Machine Learning, Data Mining, Classification)

I. INTRODUCTION

In this digital era, Oil is no longer most valuable item, most important thing is data. The value increases in many fold when it becomes transactional data. What people are buying? How frequently they are buying? What sites they do shopping? So many thing payment service provider know now just because they have transactional data. Using this knowledge, they can be more attractive to their customers. Previously, banks were very much careful about fraud transaction. Even in modern days, Banks are not less careful, but they have to use the insight from transactional data for huge competition in banking sector. Important insights can be helpful in directed advertisement, promotional offers, collaborative business with merchandiser. We studied existing works in transactional data, and using that knowledge we propose three methods to solve this problem. We will show detailed comparsion of each experiment.

II. PROBLEM DESCRIPTION

Santander is a Spanish bank. They have arrange a contest on their transactional dataset at Kaggle. The dataset is representative of original world. They have transformed all the values due to privacy issue. Given

transaction data, we should help Santander identify which customers will make a specific transaction in the future, irrespective of the amount of money transacted. They are interested in finding whether customer is happy, whether customer is interested in a particula product or loan. In the dataset they have 200K labeled training data and 200K unlabeled test data. There is a column named "target". It contains only binary values. Given transaction data, we have to predic the target column. All of the features has almost normal distribution. And target columns has an imbalanced distribution. So we have to penalize the high frequent item target in cost functon. Also We have to remove the outliers from each feature.

III. RELATED WORKS

There has been lot of works in the areas of transaction frudad analysis. Our dataset does not reveal real world feture like number of items purchased, payment amount. We did not find any paper which will solve this problem. Yet, it's a typical machine learning binary classification problem. So we studied several papers on transactional data. One of the papers [1] described removing outlier, then applying decision tree and bayesian belief network is fruitful. Another one [2,3] used ensemble methods like Random Forest and XGBoost. [4] focused on several hidden markov model strategies for Indian market. [5] did not focus on algorithm rather they focused on feature engineering. There were lot of work using deep learning. Deep learning is very successful in image and video data, for tabular data they are not much successfult. We have read one paper [6] which used auto encoder. Using the auto encoder they found important features in low dimension. This feature can be used with any prediction alogorithm.

IV. FEATURE ENGINEERING

We visualized our features to find any insight and relationship from data. All of the feature has normal distribution, It means that the algorithm should have high accuracy. Also we did boxplot. We found some outliers, We have to remove this outliers before implementing any algorithm. We also have a hit map which does not show many correlation among fetures. Feature engineering is difficult as we have 200 features. We need professional level expertise to engineer so many features.

V. PROPOSED METHODS

We have proposed several algorithms encouraged by our literature study in this report. As our dataset is tabular and feature engineering is very challenging, we took ensemble algorithms like Random Forest and Xgboost. We have also shown that how much power ful a simple deep learning based binary classifier can be. Overall we have three algorithms but each algorithm will be enriched by feature scaling, outlier removal. Finally, we will show the impact of adding binned values as extra feature. This idea is very fruitful in real world. We will also show the impact of dimensionality reduction in Random Forest.

VI. EXPERIMENT DESCRIPTION

We have proposed several algorithms in our project. Let us describe those experiments one by one. At the beginning, we tried to show how a basic Ranom Forest and Xgboost. Both of these methods shown pretty bad results. Our dataset is imbalanced, so these basic methods were producing random prediction. We tried to remove the outlier. All of our columns have normal distribution. So we can safely assume that if any value which is three standard deviation away from mean are outlier. We have also done min-max feature scaling. After that we used extra feature from each row's minimum, maximum, mean, starndard deviation, median. We also binned each column into hundred unique values and used those column as new features.

Initially, divided the training data into training and validation data Writing a random forest classifier does not take much time. But with default parameters set, we were getting very poor score. We used AUC(Area Under Curve) to measure performance. A baseline Random Forest was giving 0.5 and 0.5 for

training and validatiion. So we started hyper parameter tuning. We fixed all parameters to their default value and changed one parameter. We did this for number of trees, minimum number of samples in a leaf, max depth and class weight. From each of the graph, we got an idea. We got Test AUC at kaggle 0.83 for Random Forest. We tried some other thing to improve the perfomance for example, we used dimensionality reduction to improve performance. Unfortunately, it did not work. As features does not have higher correlation, it was very difficult to represent those points in lower dimension.

After getting promising result in Random Forest, we started working on Xgboost. Xgboost is an ensemble of multiple decision tree. But we consider the trees in sequential order. Every tree considers the weakness of previous tree and try to eradicate it. Each tree has some weight. Gradient descent is applied while goal is to minimize the loss. It is a mixture of Random forest and gradient descent. As a result, it should work better than Random Forest. It is working better than Random Forest. We got 0.89 AUC for Xgboost.

Finally, we tried to run neural network. Here hyper parameters are number of layers, number of neurons in each hidden layers, batch size, number of epochs and learning rate. We used pytorch to code it. We used binary cross entropy loss function, 256 batch size, and 100 epochs. We also used three hidden layer where first two have only 20 neurons each and last layer has 1 neuron. After everything, we applied Sigmoid. Sigmoid Converts everything into 0 to 1. Then we calculated the loss. After that, we calculated the gradient and updated all the weights. Surprisingly, with such a simple network, we 0.85 AUC. We do not have GPU. So all the processes were very slow. So we could not optimize more. But is is very good tool to give a good baseline performance.

We also did k-fold cross validation for each experiment. We saw a huge performance improvement when cross validation is used. The reason is, if we only use one set of training and validation, there will be a higher chance of error. But if we use k-fold cross validation, our training will be more robust. For each fold, we calculated the

probability and took the average to get the final probability.

VII. DISCUSSION

We learned that deep learning methods are very powerful. But for tabular data Xgboost can still be better than Deep Learning. On the other hand, Random forest is not as good as Xgboost. Feature scaling, outlier removal and binning contribute in performance improvement.

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